BLOGY AI ENGINE

Product Requirements Document

Prompt & Profit — Bizmark'26 · DTU Consulting Group × Blogy

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1. Executive Summary

This PRD defines the complete build plan for a submission to the Blogy Prompt & Profit Hackathon (Bizmark'26). The submission has three inter-linked deliverables:

- Part 1 — A scalable AI Blog Engine that converts keyword intent into SEO-optimised, GEO-ready blog content through a structured, repeatable prompt pipeline.
- Part 2 — A critical audit of the Blogy dashboard (blogy.in) covering UX, SEO, conversion, and product gaps with actionable recommendations.
- Part 3 — Two production-grade SEO blogs generated by the engine and published on five approved platforms, with live SEO validation.

All three parts are required and build on each other. Part 1 is the technical crown jewel and will be stress-tested live. Parts 2 and 3 demonstrate applied impact.

2. What We Are Building

Three distinct but connected artefacts make up the full submission:

2.1 The AI Blog Engine (Part 1)

A fully functional, end-to-end blog generation system with the following pipeline stages:

Stage	Input	Process	Output
1. Keyword Intake	Raw keyword / topic	Intent classification (informational / commercial / transactional)	Labelled keyword cluster
2. SERP Gap Analysis	Keyword cluster	Competitor top-10 scrape + gap scoring	Opportunity score + angle
3. Outline Generation	Keyword + angle	Structured H1/H2/H3 prompt with NLP rules	SEO outline with target density
4. Full Blog Draft	Outline	Section-by-section generation with tone control	~1500–2500 word draft

Stage	Input	Process	Output
5. SEO Validation	Draft	Keyword density, readability, snippet eligibility checks	SEO report + score
6. GEO Optimisation	Draft + SEO report	Add schema markup, location signals, E-E-A-T elements	GEO-ready final blog
7. Publish / Export	Final blog	Push to CMS or export Markdown/HTML	Live blog or file

2.2 Blogy Dashboard Audit (Part 2)

A structured product analysis of blogy.in covering:

- UX & technical bugs (navigation, CTA placement, mobile responsiveness, dead links)
- Conversion funnel friction — from landing page to paid plan
- SEO structural risks — indexing gaps, missing schema, thin pages
- Feature gap analysis vs. competitors (Jasper, SurferSEO, Copy.ai)
- Actionable roadmap with priority tiers (P0 / P1 / P2)

2.3 SEO Blog Publication (Part 3)

Two target blogs generated using the engine and published on 5 platforms:

- Blog A: "Blogy – Best AI Blog Automation Tool in India"
- Blog B: "How Blogy is Disrupting Martech – Organic Traffic on Autopilot, Cheapest SEO"

Target platforms: Medium, LinkedIn, WordPress.com, Substack, Hashnode

3. AI Engine Architecture (Deep Dive)

3.1 Tech Stack

Layer	Technology	Rationale
LLM Core	Anthropic Claude API (claude-sonnet-4)	Best reasoning + instruction-following for structured prompts
Keyword Research	SEMrush API / DataForSEO	Real-time SERP data, keyword volume, difficulty
SERP Scraping	SerpAPI or Bright Data	Top-10 competitor page extraction
Readability Analysis	Flesch-Kincaid (textstat Python lib)	Readability score validation
AI Detection Check	GPTZero API / Originality.ai API	AI% and naturalness score
SEO Scoring	Custom rules engine + SEMrush	Keyword density, structure, snippet check

Layer	Technology	Rationale
Frontend / Demo UI	React + Tailwind	Clean demo interface for live presentation
Backend API	Node.js / FastAPI	Prompt orchestration, pipeline controller
Publishing	Platform REST APIs (Medium, WordPress)	Auto-publish for Part 3

3.2 Prompt Architecture

The engine uses a 4-layer prompt architecture designed for clarity, repeatability, and SEO performance:

Layer 1 — System Prompt (Persona + Rules)

Sets the AI as an expert SEO content strategist. Injects global rules: keyword density 1–2%, no passive voice > 20%, Flesch readability > 60, no AI-sounding filler phrases.

Layer 2 — Context Prompt (Brand + Audience)

Injects brand voice (Blogy: confident, startup-friendly, anti-agency), target audience (Indian SME founders, growth marketers), and competitive positioning data.

Layer 3 — Task Prompt (Structured Outline)

Passes the validated SEO outline with mandatory H1/H2/H3 structure, target keywords per section, CTA placement instructions, and internal linking anchors.

Layer 4 — Validation Prompt (Self-Check)

After generation, a second Claude call reviews the draft against a 10-point SEO checklist and returns a structured JSON report with a pass/fail per metric and an overall score.

3.3 Keyword Clustering Logic

Keywords are clustered using semantic similarity + SERP overlap method:

- Step 1: Seed keyword expansion via SEMrush (top 50 related terms)
- Step 2: Group by search intent (informational / commercial / transactional)
- Step 3: SERP overlap check — keywords ranking the same top-10 URLs are clustered together
- Step 4: Pick primary keyword (highest volume × lowest difficulty) + 3–5 secondary LSI terms
- Step 5: Assign primary to H1/meta title, secondaries to H2s and body

3.4 SERP Gap Identification

For each target keyword, the engine scores the top-10 ranking pages on:

Signal	Method	Gap Threshold
Word count	Scrape + count	Target 20% more than avg. top 3
H2 coverage	Extract headings, compare	Cover angles missing from top 3

Signal	Method	Gap Threshold
Schema markup	Check structured data	Add FAQ, HowTo, Article schema if missing
Featured snippet	Check P0 box presence	Use question H2 + concise 40–60 word answer
E-E-A-T signals	Author bio, stats, citations	Add if absent in competitors

4. Blogy Dashboard Audit Plan (Part 2)

4.1 Audit Scope

Based on live review of blogy.in, the following areas will be audited:

UX & Technical Bugs

- Hero headline has a typo: "Aut mate" instead of "Automate" — critical credibility issue
- All footer links (About, Help, Terms, Documentation, Contact) are dead (#) — no destination
- "Read more..." testimonials link has no destination
- Pricing CTA buttons all route to /signin regardless of plan — no plan pre-selection
- Blog section shows "Loading blogs..." with no fallback for slow/failed fetch
- No dark mode despite modern SaaS standard
- Mobile nav not verified; sticky CTA bar absent on scroll

SEO Structural Risks

- No structured data (Schema.org) visible on homepage — missing SoftwareApplication, FAQ, Review schema
- Thin or missing meta descriptions on inner pages
- Blog section lazy-loads — risk of Google not indexing dynamically rendered content
- No sitemap.xml linked in footer or robots.txt visible
- Image alt texts using generic Unsplash photo filenames

Conversion Funnel Gaps

- "Join Waitlist" CTA competes with "Get Started" — unclear primary action
- No social proof above the fold (logo wall, press mentions)
- Pricing page has no annual/monthly toggle or comparison table vs. competitors
- No free trial or demo CTA — only "Free for first month" buried in plan card
- Onboarding flow not showcased — users don't know what happens after sign-up

4.2 Feature Gap & Innovation Ideas

Feature Idea	Priority	Impact
SEO Score Dashboard — show live domain SEO health	P0	High retention, upsell trigger

Feature Idea	Priority	Impact
Competitor Blog Spy — enter competitor URL, see their top blogs	P0	Acquisition hook
AI Humaniser Mode — toggle to reduce AI detection %	P0	Key differentiator
Content Calendar View — visual scheduling of auto-publish queue	P1	Reduces churn
GEO Targeting per Blog — set city/region for local SEO	P1	Expands TAM
WhatsApp Blog Brief — send keyword via WA, get blog	P1	India-first GTM
ROI Calculator on Pricing Page — show cost vs. agency	P2	Conversion lift
Team Collaboration + Comments on drafts	P2	Enterprise expansion

5. Blog Creation & SEO Plan (Part 3)

5.1 Blog A — Target Brief

Parameter	Value
Title	Blogy – Best AI Blog Automation Tool in India
Primary Keyword	AI blog automation tool India
Secondary Keywords	automated blogging India, SEO blog generator, AI content tool for startups
Search Intent	Commercial Investigation
Target Word Count	2,000–2,400 words
Target Readability	Flesch score > 60
Target AI Detection	< 25%
Featured Snippet Target	"What is the best AI blog tool in India?" — 50-word answer box
Schema	SoftwareApplication, FAQ, Review
CTA Placement	After intro, mid-article, and conclusion

5.2 Blog B — Target Brief

Parameter	Value
Title	How Blogy is Disrupting Martech – Organic Traffic on Autopilot, Cheapest SEO
Primary Keyword	cheapest SEO tool India
Secondary Keywords	martech disruption, organic traffic autopilot, affordable content marketing
Search Intent	Informational + Commercial
Target Word Count	1,800–2,200 words
Target Readability	Flesch score > 62
Target AI Detection	< 20%
Featured Snippet Target	"How to get organic traffic on autopilot?" — step-list format
Schema	Article, HowTo, FAQ
CTA Placement	After problem statement, before conclusion

5.3 Publishing Platform Strategy

Platform	Why Chosen	SEO Value
Medium	High DA, large tech audience, indexes fast	Backlink + referral traffic
LinkedIn Article	Decision-makers audience, startup founders	Brand authority + leads
WordPress.com	Direct Google index, structured blog format	Organic ranking potential
Substack	Newsletter + web hybrid, growing fast	Long-term subscriber capture
Hashnode	Developer + startup audience, SEO-friendly	Niche backlink, dev community

6. Success Metrics & Judging Criteria Mapping

Judging Criterion	Our Target	How We Prove It
Prompt Architecture Clarity	Documented 4-layer prompt system	Live demo + slide walkthrough
Keyword Clustering Logic	Semantic + SERP overlap method	Show clustering output in demo
SERP Gap Identification	Automated gap score per keyword	Demo gap report generation

Judging Criterion	Our Target	How We Prove It
SEO Optimisation %	> 80% on SEMrush content score	Live SEMrush screenshot
AI Detection %	< 25% on GPTZero / Originality.ai	Live detection test in demo
Snippet Readiness	> 70% probability score	Show validation output
Keyword Density Compliance	1.0–2.0% primary keyword	Density checker in UI
Scalability & Replicability	Generate new blog in < 2 min	Stress-test with new keyword
UX / Bug Identification	> 10 documented issues	Slide with annotated screenshots
Feature Innovation Depth	8 ideas with priority + impact	Prioritised roadmap table
Platform Adaptation Quality	5 published blogs, each adapted	Live links in presentation

7. Build Plan & Task Breakdown

Task	Owner	Priority	Est. Time
Set up Claude API + prompt pipeline (4 layers)	Tech	P0	4 hrs
Build keyword clustering module	Tech	P0	3 hrs
Integrate SEMrush / DataForSEO for SERP gap	Tech	P0	3 hrs
Build SEO validation + scoring engine	Tech	P0	4 hrs
Build demo UI (React)	Tech	P1	4 hrs
Run full Blogy dashboard audit	All	P0	2 hrs
Generate Blog A using engine	Tech	P0	1 hr
Generate Blog B using engine	Tech	P0	1 hr
Publish blogs on 5 platforms	All	P0	2 hrs
Validate blogs via SEMrush / GPTZero	All	P0	1 hr
Build pitch deck (5 core slides)	All	P1	3 hrs
Rehearse live demo	All	P0	2 hrs

8. Submission Checklist

- [ ] AI Blog Engine prototype — fully functional and demo-ready
- [ ] Pitch deck — max 5 core slides, saved as TeamName_P&P.pdf
- [ ] Blog A published on 5 platforms with live URLs
- [ ] Blog B published on 5 platforms with live URLs
- [ ] SEO validation screenshots (SEMrush score, AI detection %)
- [ ] Blogy audit report included in slide deck
- [ ] Live demo rehearsed with stress-test scenario

Appendix: Reference Links

- Blogy Website: https://www.blogy.in
- Blogy Dashboard: https://dashboard.blogy.in
- Hackathon: Bizmark'26 — DTU Consulting Group
- Submission Format: TeamName_P&P (.pdf or .pptx), max 5 core slides